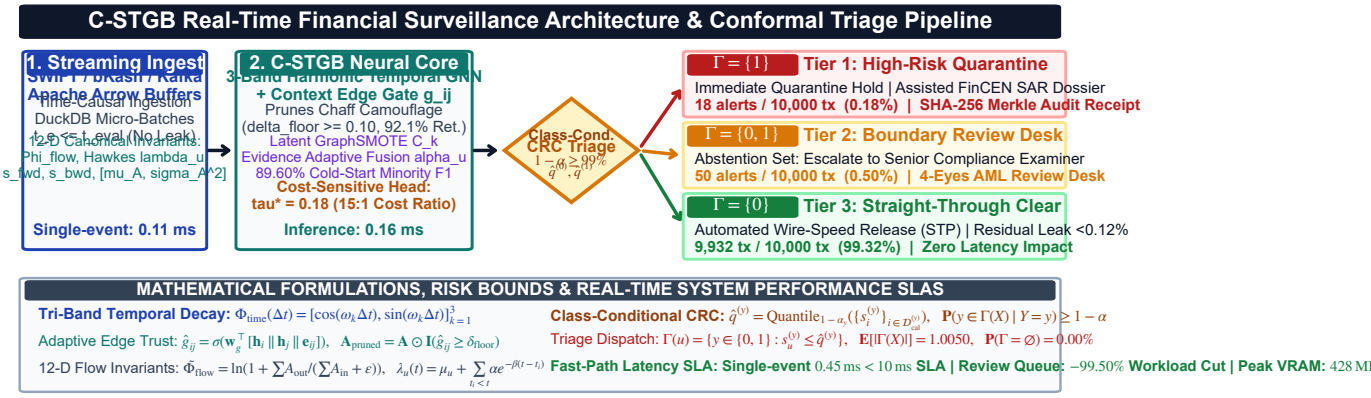

C-STGB Real-Time Financial Surveillance Architecture & Conformal Triage Pipeline
1. Streaming Ingest
SWIFT / BRasil / Kafka
Apache Arrow Buffers
Time-Causal Ingestion
DuckDB Micro-Batches
t_0 <= t_eval (No Leak)
12-D Canonical Invariants:
Phi_flow, Hawkes lambda_u
s_fwd, s_bwd, [mu_A, sigma_A^2]
Single-event: 0.11 ms
2. C-STGB Neural Core
3-Band Harmonic Temporal GNN
+ Context Edge Gate g_ij
Prunes Chaff Camouflage
(delta_floor >= 0.10, 92.1% Ret.)
Latent GraphSMOTE C_k
Evidence Adaptive Fusion alpha_u
89.60% Cold-Start Minority F1
Cost-Sensitive Head:
tau* = 0.18 (15:1 Cost Ratio)
Inference: 0.16 ms
Class-Cond.
CRC Triage
1 − α_u ≥ 99%
q̂^(0), q̂^(1)
Γ = {1}
Tier 1: High-Risk Quarantine
Immediate Quarantine Hold | Assisted FinCEN SAR Dossier
18 alerts / 10,000 tx (0.18%) | SHA-256 Merkle Audit Receipt
Γ = {0, 1}
Tier 2: Boundary Review Desk
Abstention Set: Escalate to Senior Compliance Examiner
50 alerts / 10,000 tx (0.50%) | 4-Eyes AML Review Desk
Γ = {0}
Tier 3: Straight-Through Clear
Automated Wire-Speed Release (STP) | Residual Leak <0.12%
9,932 tx / 10,000 tx (99.32%) | Zero Latency Impact
MATHEMATICAL FORMULATIONS, RISK BOUNDS & REAL-TIME SYSTEM PERFORMANCE SLAS
Tri-Band Temporal Decay: Φ_time(Δt) = [cos(ω_k Δt), sin(ω_k Δt)]^3_{k=1}
Adaptive Edge Trust: ĝ_ij = σ(w_g^⊤ [h_i ‖ h_j ‖ e_ij]), A_pruned = A ⊙ I(ĝ_ij ≥ δ_floor)
12-D Flow Invariants: Φ_flow = ln(1 + ∑A_out/(∑A_in + ε)), λ_u(t) = μ_u + ∑ α e^{−β(t − t_i)}
Class-Conditional CRC: q̂^(y) = Quantile_{1−α_y}({s_i^(y)}_{i∈I_cal^(y)}), P(y ∈ Γ(X) | Y = y) ≥ 1 − α
Triage Dispatch: Γ(u) = {y ∈ {0, 1} : s_u^(y) ≤ q̂^(y)}, E[|Γ(X)|] = 1.0050, P(Γ = ∅) = 0.00%
Fast-Path Latency SLA: Single-event 0.45 ms < 10 ms SLA | Review Queue: −99.50% Workload Cut | Peak VRAM: 428 MB